# HW8

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## Q1: 4.3

Let $\nu$ be the counting measure on $[0,1]$ and $\int_{[0,1]} h\,d\nu < \infty$ for some nonnegative function $h$. Show that $B = \{x : h(x) > 0\}$ is countable.

(**Hint:** Let $B_n = \{x : h(x) > \frac{1}{n}\}$. Show that $B_n$ is a finite set for each $n \in \mathbb{N}$.)

### Answer

We utilize the following results:

- Counting measure: $\nu(A)$ is the number of elements of $A$; so $\nu(A) < \infty$ iff $A$ is finite.
- Corollary 2.3.3: Proposition 2.3.1 holds for nonnegative measurable functions; in particular, $\int \alpha f\,d\mu = \alpha \int f\,d\mu$ for $\alpha \geq 0$, and $f \geq g$ a.e. $(\mu)$ implies $\int f\,d\mu \geq \int g\,d\mu$.
- Proposition A.2.1 (Axiom of Archimedes): for every $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ with $n > y$.

*Proof.* ($h$ is taken to be measurable, as the integral presumes.) For $n \in \mathbb{N}$, define $B_n \equiv \{x \in [0,1] : h(x) > \frac{1}{n}\}$, a measurable set since $h$ is measurable.

**$B_n$ is finite:** Let $n \in \mathbb{N}$ be given. On $B_n$, $h > \frac{1}{n}$, and off $B_n$, $h \geq 0$, so $h \geq \frac{1}{n} I_{B_n}$ on $[0,1]$. By Corollary 2.3.3,

$$\frac{1}{n}\,\nu(B_n) = \int \frac{1}{n} I_{B_n}\,d\nu \leq \int_{[0,1]} h\,d\nu < \infty.$$

Thus $\nu(B_n) < \infty$, i.e., $B_n$ is finite.

**$B$ is countable:** If $x \in B$, then $h(x) > 0$, and by Proposition A.2.1 there is an $n \in \mathbb{N}$ with $n > 1/h(x)$, i.e., $h(x) > \frac{1}{n}$, so $x \in B_n$. Conversely, each $B_n \subset B$. So $B = \bigcup_{n \geq 1} B_n$, a countable union of finite sets, and thus countable.

Said differently, this is the fact that makes Remark 4.1.1 work, where it shows the Radon-Nikodym theorem can fail when $\nu$ is not $\sigma$-finite. //

## Q2: 4.4

Prove Proposition 4.1.2.

Proposition 4.1.2: Let $\nu, \mu, \mu_1, \mu_2, \ldots$ be $\sigma$-finite measures on a measurable space $(\Omega, \mathcal{F})$.

(i) If $\mu_1 \ll \mu_2$ and $\mu_2 \ll \mu_3$, then $\mu_1 \ll \mu_3$ and

$$\frac{d\mu_1}{d\mu_3} = \frac{d\mu_1}{d\mu_2}\frac{d\mu_2}{d\mu_3} \quad \text{a.e. } (\mu_3).$$

(ii) Suppose that $\mu_1$ and $\mu_2$ are dominated by $\mu_3$. Then for any $\alpha, \beta \geq 0$, $\alpha\mu_1 + \beta\mu_2$ is dominated by $\mu_3$ and

$$\frac{d(\alpha\mu_1 + \beta\mu_2)}{d\mu_3} = \alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3} \quad \text{a.e. } (\mu_3).$$

(iii) If $\mu \ll \nu$ and $\frac{d\mu}{d\nu} > 0$ a.e. $(\nu)$, then $\nu \ll \mu$ and

$$\frac{d\nu}{d\mu} = \left(\frac{d\mu}{d\nu}\right)^{-1} \quad \text{a.e. } (\mu).$$

(iv) Let $\{\mu_n\}_{n\geq 1}$ be a sequence of measures and $\{\alpha_n\}_{n\geq 1}$ be a sequence of positive real numbers, i.e., $\alpha_n > 0$ for all $n \geq 1$. Define $\mu = \sum_{n=1}^{\infty} \alpha_n \mu_n$.

(a) Then, $\mu \ll \nu$ iff $\mu_n \ll \nu$ for each $n \geq 1$ and in this case,

$$\frac{d\mu}{d\nu} = \sum_{n=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu} \quad \text{a.e. } (\nu).$$

(b) $\mu \perp \nu$ iff $\mu_n \perp \nu$ for all $n \geq 1$.

## Answer

We utilize the following results:

- Definition 4.1.1 (absolute continuity): $\mu \ll \nu$ if $\nu(A) = 0 \Rightarrow \mu(A) = 0$ for all $A \in \mathcal{F}$.
- Definition 4.1.2 (singularity): $\mu \perp \nu$ if there is a $B \in \mathcal{F}$ with $\mu(B) = 0$ and $\nu(B^c) = 0$.
- Definition 4.1.3 (Radon-Nikodym derivative): a nonnegative measurable $h$ with $\mu(A) = \int_A h\, d\nu$ for all $A \in \mathcal{F}$ is written $\frac{d\mu}{d\nu} = h$; when $\mu$ is $\sigma$-finite, it is unique a.e. $(\nu)$.
- Example 4.1.3: if $h \geq 0$ is measurable and $\mu(A) \equiv \int_A h\, d\nu$, then $\mu$ is a measure and $\mu \ll \nu$.
- Theorem 4.1.1 (Lebesgue-Radon-Nikodym): for $\sigma$-finite $\mu, \nu$, (i) $\mu = \mu_a + \mu_s$ uniquely, with $\mu_a \ll \nu$ and $\mu_s \perp \nu$; (ii) there is a nonnegative measurable $h$ with $\mu_a(A) = \int_A h\, d\nu$ for all $A \in \mathcal{F}$. In particular, if $\mu \ll \nu$, then $\mu_a = \mu$, and $\frac{d\mu}{d\nu}$ exists.
- Definitions 2.3.2 and 2.3.3 (integrals of simple and of nonnegative measurable functions), with nonnegative simple $h_n \uparrow h$ as in (3.7).
- Corollary 2.3.3 (see Q1); in particular, $\int(\alpha f + \beta g)\, d\mu = \alpha \int f\, d\mu + \beta \int g\, d\mu$ for $\alpha, \beta \geq 0$.
- Theorem 2.3.4 (monotone convergence theorem, MCT): for nonnegative measurable $f_n \uparrow f$, $\int f_n\, d\mu \uparrow \int f\, d\mu$.
- Corollary 2.3.5: for nonnegative measurable $\{h_n\}_{n\geq 1}$, $\int \left(\sum_{n=1}^{\infty} h_n\right) d\mu = \sum_{n=1}^{\infty} \int h_n\, d\mu$.
- Problem 2.20: for nonnegative $\{a_{ij}\}$, $\sum_i \sum_j a_{ij} = \sum_j \sum_i a_{ij}$.

*Proof.*

**Lemma:** Let $\mu, \nu$ be measures on $(\Omega, \mathcal{F})$, and let $f \geq 0$ be measurable with $\mu(A) = \int_A f\, d\nu$ for all $A \in \mathcal{F}$. Then for every nonnegative measurable $h$,

$$\int h\, d\mu = \int hf\, d\nu. \tag{1}$$

We build up from indicators.

*Indicator $h$:* If $h = I_A$, $A \in \mathcal{F}$, then $\int I_A\, d\mu = \mu(A) = \int_A f\, d\nu = \int I_A f\, d\nu$.

*Nonnegative simple $h$:* If $h = \sum_{i=1}^{k} c_i I_{A_i}$ with $c_i \geq 0$, then by Corollary 2.3.3 and the indicator case,

$$\int h\, d\mu = \sum_{i=1}^{k} c_i \int I_{A_i}\, d\mu = \sum_{i=1}^{k} c_i \int I_{A_i} f\, d\nu = \int hf\, d\nu.$$

*Nonnegative measurable $h$:* Let $\{h_n\}_{n\geq 1}$ be nonnegative simple functions with $h_n \uparrow h$, as in (3.7). Then $h_n f \uparrow hf$, and by Definition 2.3.3, the simple case, and the MCT,

$$\int h\, d\mu = \lim_{n\to\infty} \int h_n\, d\mu = \lim_{n\to\infty} \int h_n f\, d\nu = \int hf\, d\nu.$$

Throughout, each Radon-Nikodym derivative used exists by Theorem 4.1.1, since all measures involved are $\sigma$-finite.

## (i)

**Answer**

*Proof.* $\mu_1 \ll \mu_3$**:** Suppose $\mu_3(A) = 0$ for some $A \in \mathcal{F}$. Then $\mu_2(A) = 0$, since $\mu_2 \ll \mu_3$, and so $\mu_1(A) = 0$, since $\mu_1 \ll \mu_2$. Thus $\mu_1 \ll \mu_3$.

**The chain rule:** Let $A \in \mathcal{F}$ be given. Applying (1) to $\mu_2$ and $\mu_3$, with $f = \frac{d\mu_2}{d\mu_3}$ and $h = I_A \frac{d\mu_1}{d\mu_2}$,

$$\mu_1(A) = \int I_A \frac{d\mu_1}{d\mu_2}\, d\mu_2 = \int I_A \frac{d\mu_1}{d\mu_2}\frac{d\mu_2}{d\mu_3}\, d\mu_3 = \int_A \frac{d\mu_1}{d\mu_2}\frac{d\mu_2}{d\mu_3}\, d\mu_3.$$

So $\frac{d\mu_1}{d\mu_2}\frac{d\mu_2}{d\mu_3}$ is a Radon-Nikodym derivative of $\mu_1$ w.r.t. $\mu_3$, and by uniqueness (Definition 4.1.3), $\frac{d\mu_1}{d\mu_3} = \frac{d\mu_1}{d\mu_2}\frac{d\mu_2}{d\mu_3}$ a.e. $(\mu_3)$. //

## (ii)

**Answer**

*Proof.* First, $\alpha\mu_1 + \beta\mu_2$ is a $\sigma$-finite measure. It is a measure since $\alpha, \beta \geq 0$. For $\sigma$-finiteness, take partitions $\{D_n\}$ and $\{E_m\}$ of $\Omega$ with $\mu_1(D_n) < \infty$ and $\mu_2(E_m) < \infty$. Then $\{D_n \cap E_m\}_{n,m\geq 1}$ is a countable partition of $\Omega$ on which $\alpha\mu_1 + \beta\mu_2$ is finite.

$\alpha\mu_1 + \beta\mu_2 \ll \mu_3$**:** Suppose $\mu_3(A) = 0$. Then $\mu_1(A) = \mu_2(A) = 0$, so $(\alpha\mu_1 + \beta\mu_2)(A) = 0$.

**The derivative:** For $A \in \mathcal{F}$, by Corollary 2.3.3,

$$(\alpha\mu_1 + \beta\mu_2)(A) = \alpha \int_A \frac{d\mu_1}{d\mu_3}\, d\mu_3 + \beta \int_A \frac{d\mu_2}{d\mu_3}\, d\mu_3 = \int_A \left(\alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3}\right) d\mu_3.$$

Since $\alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3}$ is nonnegative and measurable, it is a Radon-Nikodym derivative of $\alpha\mu_1 + \beta\mu_2$ w.r.t. $\mu_3$, and by uniqueness the identity holds a.e. $(\mu_3)$. //

## (iii)

**Answer**

*Proof.* Write $f \equiv \frac{d\mu}{d\nu}$ and $N \equiv \{f = 0\}$, so that $\nu(N) = 0$. Define $g \equiv \frac{1}{f} I_{N^c}$, which is nonnegative and measurable (as in Problem 2.3), with $gf = I_{N^c}$.

Let $A \in \mathcal{F}$ be given. Since $\nu(N) = 0$, and by (1) with $h = I_A g$,

$$\nu(A) = \nu(A \cap N^c) = \int I_A\, gf\, d\nu = \int I_A\, g\, d\mu = \int_A g\, d\mu.$$

Thus $\nu \ll \mu$ (Example 4.1.3) and $\frac{d\nu}{d\mu} = g$ a.e. $(\mu)$ by uniqueness. Finally, $\mu(N) = \int_N f\, d\nu = 0$, since $f = 0$ on $N$. So $g = \frac{1}{f}$ a.e. $(\mu)$, i.e., $\frac{d\nu}{d\mu} = \left(\frac{d\mu}{d\nu}\right)^{-1}$ a.e. $(\mu)$. //

## (iv)

**Answer**

*Proof.* First, $\mu$ is a measure. Clearly $\mu(\emptyset) = 0$ and $\mu \geq 0$. For disjoint $A_1, A_2, \ldots \in \mathcal{F}$, by Problem 2.20,

$$\mu\Big(\bigcup_{k\geq 1} A_k\Big) = \sum_{n=1}^{\infty} \alpha_n \sum_{k=1}^{\infty} \mu_n(A_k) = \sum_{k=1}^{\infty}\sum_{n=1}^{\infty} \alpha_n \mu_n(A_k) = \sum_{k=1}^{\infty} \mu(A_k).$$

For any $A \in \mathcal{F}$, since every $\alpha_n > 0$ and every $\mu_n(A) \geq 0$,

$$\mu(A) = 0 \iff \mu_n(A) = 0 \text{ for all } n \geq 1. \tag{2}$$

**(a):**

($\Rightarrow$) Suppose $\mu \ll \nu$, and let $\nu(A) = 0$. Then $\mu(A) = 0$, so $\mu_n(A) = 0$ for all $n$ by (2). Thus $\mu_n \ll \nu$ for each $n \geq 1$.

($\Leftarrow$) Suppose $\mu_n \ll \nu$ for each $n \geq 1$, and let $\nu(A) = 0$. Then $\mu_n(A) = 0$ for all $n$, so $\mu(A) = 0$ by (2). Thus $\mu \ll \nu$.

**The derivative:** In this case, for $A \in \mathcal{F}$, Corollary 2.3.5 gives

$$\mu(A) = \sum_{n=1}^{\infty} \alpha_n \int_A \frac{d\mu_n}{d\nu}\, d\nu = \int_A \sum_{n=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu}\, d\nu.$$

Since $\sum_{n=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu}$ is nonnegative and measurable, it is a Radon-Nikodym derivative of $\mu$ w.r.t. $\nu$, and by uniqueness the identity holds a.e. ($\nu$).

**(b):**

($\Rightarrow$) Suppose $\mu \perp \nu$, with $B \in \mathcal{F}$ such that $\mu(B) = 0$ and $\nu(B^c) = 0$. By (2), $\mu_n(B) = 0$ for all $n$, so the same $B$ gives $\mu_n \perp \nu$ for all $n \geq 1$.

($\Leftarrow$) Suppose $\mu_n \perp \nu$ for all $n \geq 1$, with $B_n \in \mathcal{F}$ such that $\mu_n(B_n) = 0$ and $\nu(B_n^c) = 0$. Define $D \equiv \bigcap_{n\geq 1} B_n \in \mathcal{F}$. Since $D \subset B_n$, $\mu_n(D) = 0$ for all $n$, so $\mu(D) = 0$ by (2). By countable subadditivity,

$$\nu(D^c) = \nu\Big(\bigcup_{n\geq 1} B_n^c\Big) \leq \sum_{n=1}^{\infty} \nu(B_n^c) = 0.$$

Thus $\mu \perp \nu$. //

# Q3: 4.5

Find the Lebesgue decomposition of $\mu$ w.r.t. $\nu$ and the Radon-Nikodym derivative $\frac{d\mu_a}{d\nu}$ in the following cases where $\mu_a$ is the absolutely continuous component of $\mu$ w.r.t. $\nu$.

(a) $\mu = N(0,1)$, $\nu = \text{Exponential}(1)$

(b) $\mu = \text{Exponential}(1)$, $\nu = N(0,1)$

(c) $\mu = \mu_1 + \mu_2$, where $\mu_1 = N(0,1)$, $\mu_2 = \text{Poisson}(1)$ and $\nu = \text{Cauchy}(0,1)$.

(d) $\mu = \mu_1 + \mu_2$, $\nu = \text{Geometric}(p)$, $0 < p < 1$, where $\mu_1 = N(0,1)$ and $\mu_2 = \text{Poisson}(1)$.

(e) $\mu = \mu_1 + \mu_2$, $\nu = \nu_1 + \nu_2$ where $\mu_1 = N(0,1)$, $\mu_2 = \text{Poisson}(1)$, $\nu_1 = \text{Cauchy}(0,1)$ and $\nu_2 = \text{Geometric}(p)$, $0 < p < 1$.

(f) $\mu = \text{Binomial}(10, 1/2)$, $\nu = \text{Poisson}(1)$.

The measures referred to above are defined in Tables 4.6.1 and 4.6.2, given at the end of this section.

## Answer

Note: Table 4.6.1 prints Geometric($p$) as $\sum_{i=0}^{\infty} p(1-p)^{i-1} I_A(i)$, which does not sum to 1. We take Geometric($p$) on $\mathbb{N} = \{1, 2, \ldots\}$ with mass $p(1-p)^{i-1}$ at $i$. In particular, Geometric($p$) puts no mass at 0.

We utilize the following results:

- Definitions 4.1.1–4.1.3, Example 4.1.3, and Theorem 4.1.1 (see Q2).
- Proposition 4.1.2 (see Q2), and the Lemma (1).
- Corollary 2.3.3 (see Q1).

Write $m$ for Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $\kappa$ for the counting measure on $\mathbb{Z}_+ = \{0, 1, 2, \ldots\}$, i.e., $\kappa(A) \equiv \#(A \cap \mathbb{Z}_+)$, so that $\int_A h \, d\kappa = \sum_{k \in A \cap \mathbb{Z}_+} h(k)$ for nonnegative $h$ (as in Remark 2.3.2). From Tables 4.6.1 and 4.6.2, the measures here have the following densities.

| Measure | Density | w.r.t. |
|---|---|---|
| $N(0,1)$ | $\phi(x) \equiv \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ | $m$ |
| Exponential(1) | $e^{-x} I_{(0,\infty)}(x)$ | $m$ |
| Cauchy$(0,1)$ | $c(x) \equiv \frac{1}{\pi(1+x^2)}$ | $m$ |
| Poisson$(\lambda)$ | $e^{-\lambda} \frac{\lambda^x}{x!} I_{\mathbb{Z}_+}(x)$ | $\kappa$ |
| Geometric$(p)$ | $p(1-p)^{x-1} I_{\mathbb{N}}(x)$ | $\kappa$ |
| Binomial$(10, 1/2)$ | $\binom{10}{x} 2^{-10} I_{\{0,1,\ldots,10\}}(x)$ | $\kappa$ |

Any measure with a density w.r.t. $m$ vanishes on countable sets, since it is $\ll m$ (Example 4.1.3) and $m$ of a countable set is 0.

**Claim:** Let $\lambda$ be a measure, and let $\mu$ and $\nu$ be finite measures with $\mu(A) = \int_A f \, d\lambda$ and $\nu(A) = \int_A g \, d\lambda$ for all $A$, for nonnegative measurable $f, g$. Let $B \equiv \{g > 0\}$. Then the Lebesgue decomposition of $\mu$ w.r.t. $\nu$ is

$$\mu_a(A) = \mu(A \cap B), \quad \mu_s(A) = \mu(A \cap B^c), \quad \text{with} \quad \frac{d\mu_a}{d\nu} = \frac{f}{g} I_B \ \text{ a.e. } (\nu). \tag{3}$$

$\mu_a$ and $\mu_s$ are finite measures (restrictions of $\mu$) with $\mu = \mu_a + \mu_s$. $\mu_s \perp \nu$, since $\mu_s(B) = 0$ and $\nu(B^c) = \int_{B^c} g \, d\lambda = 0$, as $g = 0$ on $B^c$. Next, $\frac{f}{g} I_B$ is nonnegative and measurable (as in Problem 2.3), and $\frac{f}{g} I_B \cdot g = f I_B$. So by (1), applied to $\nu$ and $\lambda$ with $h = I_A \frac{f}{g} I_B$,

$$\int_A \frac{f}{g} I_B \, d\nu = \int I_A \frac{f}{g} I_B \, g \, d\lambda = \int_{A \cap B} f \, d\lambda = \mu_a(A).$$

Thus $\mu_a \ll \nu$ (Example 4.1.3) with $\frac{d\mu_a}{d\nu} = \frac{f}{g} I_B$. By the uniqueness in Theorem 4.1.1(i), $(\mu_a, \mu_s)$ is the Lebesgue decomposition.

## (a)

**Answer**

$\mu_a(A) = \mu(A \cap (0, \infty))$, $\mu_s(A) = \mu(A \cap (-\infty, 0])$, and

$$\frac{d\mu_a}{d\nu}(x) = \frac{1}{\sqrt{2\pi}} e^{x - x^2/2} I_{(0,\infty)}(x) \quad \text{a.e. } (\nu).$$

*Proof.* Apply (3) with $\lambda = m$, $f = \phi$, and $g(x) = e^{-x} I_{(0,\infty)}(x)$, so that $B = (0, \infty)$ and $\frac{f}{g} I_B = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} e^x I_{(0,\infty)}$. //

## (b)

**Answer**

$\mu_a = \mu$ and $\mu_s = 0$, i.e., $\mu \ll \nu$, and

$$\frac{d\mu}{d\nu}(x) = \sqrt{2\pi}\, e^{x^2/2 - x} I_{(0,\infty)}(x) \quad \text{a.e. } (\nu).$$

*Proof.* Apply (3) with $\lambda = m$, $f(x) = e^{-x} I_{(0,\infty)}(x)$, and $g = \phi > 0$, so that $B = \mathbb{R}$, $\mu_s(A) = \mu(A \cap \emptyset) = 0$, and $\frac{f}{g} = \sqrt{2\pi}\, e^{x^2/2} e^{-x} I_{(0,\infty)}$. //

## (c)

**Answer**

$\mu_a = \mu_1$ and $\mu_s = \mu_2$, and

$$\frac{d\mu_a}{d\nu}(x) = \sqrt{\frac{\pi}{2}}\,(1 + x^2)\, e^{-x^2/2} \quad \text{a.e. } (\nu).$$

*Proof.* $\mu_1 \ll \nu$**:** Apply (3) to $\mu_1$ and $\nu$ with $\lambda = m$, $f = \phi$, and $g = c > 0$. Then $B = \mathbb{R}$, so $\mu_1 \ll \nu$ with $\frac{d\mu_1}{d\nu} = \frac{\phi}{c} = \sqrt{\frac{\pi}{2}}(1 + x^2)e^{-x^2/2}$.

$\mu_2 \perp \nu$**:** Let $D \equiv \mathbb{R} \setminus \mathbb{Z}_+$. Then $\mu_2(D) = 0$, and $\nu(D^c) = \nu(\mathbb{Z}_+) = 0$, since $\mathbb{Z}_+$ is countable and $\nu$ has a density w.r.t. $m$.

Putting this all together, $\mu = \mu_1 + \mu_2$ with $\mu_1 \ll \nu$ and $\mu_2 \perp \nu$, so by the uniqueness in Theorem 4.1.1(i), $\mu_a = \mu_1$, $\mu_s = \mu_2$, and $\frac{d\mu_a}{d\nu} = \frac{d\mu_1}{d\nu}$. //

## (d)

**Answer**

$\mu_a(A) = \mu_2(A \cap \mathbb{N})$, $\mu_s(A) = \mu_1(A) + \mu_2(A \cap \{0\})$, and

$$\frac{d\mu_a}{d\nu}(x) = \frac{(1-p)^{1-x}}{e\, p\, x!} I_{\mathbb{N}}(x) \quad \text{a.e. } (\nu).$$

*Proof.* $\mu_2$ **against** $\nu$**:** Apply (3) to $\mu_2$ and $\nu$ with $\lambda = \kappa$, $f(x) = \frac{e^{-1}}{x!} I_{\mathbb{Z}_+}(x)$, and $g(x) = p(1-p)^{x-1} I_{\mathbb{N}}(x)$, so that $B = \mathbb{N}$. Then $\mu_2 = \mu_{2a} + \mu_{2s}$, where $\mu_{2a}(A) \equiv \mu_2(A \cap \mathbb{N}) \ll \nu$ and $\mu_{2s}(A) \equiv \mu_2(A \cap \mathbb{N}^c) = \mu_2(A \cap \{0\})$, as $\mu_2$ lives on $\mathbb{Z}_+$. Also

$$\frac{d\mu_{2a}}{d\nu}(x) = \frac{e^{-1}/x!}{p(1-p)^{x-1}} I_{\mathbb{N}}(x) = \frac{(1-p)^{1-x}}{e\, p\, x!} I_{\mathbb{N}}(x).$$

$\mu_1 + \mu_{2s} \perp \nu$**:** Take $D \equiv \mathbb{N}$. Then $\mu_1(\mathbb{N}) = 0$, since $\mathbb{N}$ is countable and $\mu_1$ has a density w.r.t. $m$, and $\mu_{2s}(\mathbb{N}) = \mu_2(\mathbb{N} \cap \{0\}) = 0$. Also $\nu(\mathbb{N}^c) = 0$.

Putting this all together, $\mu = \mu_{2a} + (\mu_1 + \mu_{2s})$ with $\mu_{2a} \ll \nu$ and $\mu_1 + \mu_{2s} \perp \nu$, so by uniqueness, $\mu_a = \mu_{2a}$ and $\mu_s = \mu_1 + \mu_{2s}$. //

## (e)

**Answer**

$\mu_a(A) = \mu_1(A) + \mu_2(A \cap \mathbb{N})$, $\mu_s(A) = \mu_2(A \cap \{0\})$, and

$$\frac{d\mu_a}{d\nu}(x) = \sqrt{\frac{\pi}{2}}\,(1 + x^2)\, e^{-x^2/2} I_{\mathbb{N}^c}(x) + \frac{(1-p)^{1-x}}{e\, p\, x!} I_{\mathbb{N}}(x) \quad \text{a.e. } (\nu).$$

*Proof.* Let $\mu_{2a}$ and $\mu_{2s}$ be as in (d). The tricky part is that $\nu$ adds mass on $\mathbb{N}$, so $\frac{d\mu_1}{d\nu}$ is not $\frac{d\mu_1}{d\nu_1}$ everywhere. We pass through $\nu_1$ and $\nu_2$.

$\nu_1, \nu_2 \ll \nu$**:** Note that $\nu_1(\mathbb{N}) = 0$, since $\nu_1$ has a density w.r.t. $m$, and $\nu_2(\mathbb{N}^c) = 0$. So for $A \in \mathcal{B}(\mathbb{R})$,

$$\nu_1(A) = \nu_1(A \cap \mathbb{N}^c) = \nu(A \cap \mathbb{N}^c) = \int_A I_{\mathbb{N}^c}\, d\nu \quad \text{and} \quad \nu_2(A) = \nu_2(A \cap \mathbb{N}) = \nu(A \cap \mathbb{N}) = \int_A I_{\mathbb{N}}\, d\nu.$$

Thus $\nu_1, \nu_2 \ll \nu$, with $\frac{d\nu_1}{d\nu} = I_{\mathbb{N}^c}$ and $\frac{d\nu_2}{d\nu} = I_{\mathbb{N}}$.

$\mu_1 \ll \nu$**:** By (c), $\mu_1 \ll \nu_1$ with $\frac{d\mu_1}{d\nu_1} = \frac{\phi}{c}$. By Proposition 4.1.2(i),

$$\frac{d\mu_1}{d\nu} = \frac{d\mu_1}{d\nu_1}\frac{d\nu_1}{d\nu} = \sqrt{\frac{\pi}{2}}(1+x^2)e^{-x^2/2}I_{\mathbb{N}^c}(x) \quad \text{a.e. } (\nu).$$

$\mu_{2a} \ll \nu$**:** By (d), with $\nu_2$ in the role of $\nu$, $\mu_{2a} \ll \nu_2$ with $\frac{d\mu_{2a}}{d\nu_2} = \frac{(1-p)^{1-x}}{e\,p\,x!}I_{\mathbb{N}}$. By Proposition 4.1.2(i),

$$\frac{d\mu_{2a}}{d\nu} = \frac{d\mu_{2a}}{d\nu_2}\frac{d\nu_2}{d\nu} = \frac{(1-p)^{1-x}}{e\,p\,x!}I_{\mathbb{N}}(x) \quad \text{a.e. } (\nu).$$

$\mu_{2s} \perp \nu$**:** Let $D \equiv \mathbb{R} \setminus \{0\}$. Then $\mu_{2s}(D) = 0$, and $\nu(\{0\}) = \nu_1(\{0\}) + \nu_2(\{0\}) = 0 + 0 = 0$.

Putting this all together, $\mu = (\mu_1 + \mu_{2a}) + \mu_{2s}$, where $\mu_1 + \mu_{2a} \ll \nu$ by Proposition 4.1.2(ii) and $\mu_{2s} \perp \nu$. By uniqueness, $\mu_a = \mu_1 + \mu_{2a}$ and $\mu_s = \mu_{2s}$, and Proposition 4.1.2(ii) gives $\frac{d\mu_a}{d\nu} = \frac{d\mu_1}{d\nu} + \frac{d\mu_{2a}}{d\nu}$ a.e. $(\nu)$. //

## (f)

**Answer**

$\mu_a = \mu$ and $\mu_s = 0$, i.e., $\mu \ll \nu$, and

$$\frac{d\mu}{d\nu}(x) = \binom{10}{x}\frac{e\,x!}{2^{10}}I_{\{0,\ldots,10\}}(x) = \frac{e \cdot 10!}{2^{10}\,(10-x)!}I_{\{0,\ldots,10\}}(x) \quad \text{a.e. } (\nu).$$

*Proof.* Apply (3) with $\lambda = \kappa$, $f(x) = \binom{10}{x}2^{-10}I_{\{0,\ldots,10\}}(x)$, and $g(x) = \frac{e^{-1}}{x!}I_{\mathbb{Z}_+}(x)$. Then $B = \mathbb{Z}_+ \supset \{0,\ldots,10\}$, so $\mu_s(A) = \mu(A \cap \mathbb{Z}_+^c) = 0$, and $\frac{f}{g}I_B$ is as displayed, using $\binom{10}{x}x! = \frac{10!}{(10-x)!}$. //

# Q4: Extra A

Let $(\Omega, \mathcal{F})$ be a measurable space, and $\mu, \mu_1, \mu_2, \nu$ are $\sigma$-finite measures on it. Define the set function $\mu_3 \equiv \mu_1 + \mu_2$ (i.e., for all $A \in \mathcal{F}$, $\mu_3(A) = \mu_1(A) + \mu_2(A)$). Then prove the following:

(a) $\mu_3$ is a measure

(b) if $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$

(c) Express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ (proof required).

(d) If $\mu \ll \nu$ and $\mu \perp \nu$ holds, show that $\mu$ is the zero measure.

(e) If $\mu_1 = \text{Lognormal}(5,6)$, $\mu_2 = \text{Geometric}(0.3)$, and $\nu = \text{Poisson}(8)$ on $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then find the Lebesgue decomposition of $\mu_3$ w.r.t. $\nu$, and find the Radon-Nikodym derivative $\frac{d\mu_{3a}}{d\nu}$, where $\mu_{3a}$ is the absolutely continuous component of $\mu_3$ in the decomposition.

We utilize the following results for (a)–(e):

- Definitions 4.1.1–4.1.3, Example 4.1.3, and Theorem 4.1.1 (see Q2).
- Proposition 4.1.2(ii) (see Q2).
- Corollary 2.3.3 (see Q1).
- The Claim (3) and the densities tabled in Q3.

## (a)

**Answer**

*Proof.* We verify the properties of a measure.

$\mu_3 \geq 0$: For $A \in \mathcal{F}$, $\mu_1(A), \mu_2(A) \in [0, \infty]$, so $\mu_3(A) \in [0, \infty]$.

$\mu_3(\emptyset) = 0$: $\mu_3(\emptyset) = \mu_1(\emptyset) + \mu_2(\emptyset) = 0 + 0 = 0$.

Countable additivity: Let $A_1, A_2, \ldots \in \mathcal{F}$ be disjoint. Then

$$\mu_3\Big(\bigcup_{n\geq 1} A_n\Big) = \sum_{n=1}^{\infty} \mu_1(A_n) + \sum_{n=1}^{\infty} \mu_2(A_n) = \sum_{n=1}^{\infty} \big(\mu_1(A_n) + \mu_2(A_n)\big) = \sum_{n=1}^{\infty} \mu_3(A_n),$$

where the series of nonnegative terms may be added termwise. Thus $\mu_3$ is a measure. //

## (b)

**Answer**

*Proof.* Let $A \in \mathcal{F}$ with $\nu(A) = 0$ be given. Since $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, $\mu_1(A) = \mu_2(A) = 0$. Thus $\mu_3(A) = 0 + 0 = 0$, and $\mu_3 \ll \nu$. //

## (c)

**Answer**

$\frac{d\mu_3}{d\nu} = \frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}$ a.e. $(\nu)$.

*Proof.* (This is Q2 (ii) with $\alpha = \beta = 1$.) The derivatives $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ exist by Theorem 4.1.1, since $\mu_1, \mu_2 \ll \nu$, all $\sigma$-finite. For $A \in \mathcal{F}$, by Corollary 2.3.3,

$$\mu_3(A) = \int_A \frac{d\mu_1}{d\nu}\, d\nu + \int_A \frac{d\mu_2}{d\nu}\, d\nu = \int_A \left(\frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}\right) d\nu.$$

So $\frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}$, being nonnegative and measurable, is a Radon-Nikodym derivative of $\mu_3$ w.r.t. $\nu$. By uniqueness (Definition 4.1.3), the identity holds a.e. $(\nu)$. //

## (d)

**Answer**

*Proof.* Since $\mu \perp \nu$, there is a $B \in \mathcal{F}$ with $\mu(B) = 0$ and $\nu(B^c) = 0$. Since $\mu \ll \nu$, $\nu(B^c) = 0$ gives $\mu(B^c) = 0$. So

$$\mu(\Omega) = \mu(B) + \mu(B^c) = 0,$$

and for any $A \in \mathcal{F}$, $0 \leq \mu(A) \leq \mu(\Omega) = 0$. Thus $\mu$ is the zero measure. //

## (e)

**Answer**

$\mu_{3a} = \mu_2$ and $\mu_{3s} = \mu_1$, and

$$\frac{d\mu_{3a}}{d\nu}(x) = \frac{0.3\,(0.7)^{x-1}}{e^{-8}\,8^x/x!} I_{\mathbb{N}}(x) = \frac{3}{7}\, e^8\, x! \left(\frac{7}{80}\right)^x I_{\mathbb{N}}(x) \quad \text{a.e. } (\nu).$$

(As in Q3, Geometric(0.3) lives on $\mathbb{N}$, and Lognormal$(5, 6)$ has $\gamma = 5$, $\sigma^2 = 6$ in Table 4.6.2.)

*Proof.* $\mu_2 \ll \nu$**:** Apply (3) to $\mu_2$ and $\nu$ with $\lambda = \kappa$, $f(x) = 0.3\,(0.7)^{x-1} I_{\mathbb{N}}(x)$, and $g(x) = e^{-8}\frac{8^x}{x!} I_{\mathbb{Z}_+}(x)$. Then $B = \mathbb{Z}_+ \supset \mathbb{N}$, so the singular part $\mu_2(A \cap \mathbb{Z}_+^c)$ is 0. Thus $\mu_2 \ll \nu$, with $\frac{d\mu_2}{d\nu} = \frac{f}{g} I_{\mathbb{Z}_+}$ as displayed, using $0.3\,(0.7)^{x-1}/8^x = \frac{3}{7}\left(\frac{7}{80}\right)^x$.

$\mu_1 \perp \nu$**:** Let $D \equiv \mathbb{Z}_+$. Then $\mu_1(\mathbb{Z}_+) = 0$, since $\mathbb{Z}_+$ is countable and $\mu_1$ has a density w.r.t. $m$, and $\nu(\mathbb{Z}_+^c) = 0$.

Putting this all together, $\mu_3 = \mu_2 + \mu_1$ with $\mu_2 \ll \nu$ and $\mu_1 \perp \nu$, so by the uniqueness in Theorem 4.1.1(i), $\mu_{3a} = \mu_2$ and $\mu_{3s} = \mu_1$. //